

Zijiang Yan

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Summary

Cloud & AI Platform Engineer with 5+ years of experience designing and operating scalable *DevOps* and cloud infrastructure across **AWS and Azure** in enterprise environments. Strong expertise in **Kubernetes, CI/CD automation**, and **Infrastructure-as-Code (Terraform)**, with a focus on *cloud-native platform engineering* for distributed systems. Hands-on experience building **AI/ML platforms**, *data pipelines*, and *model deployment workflows*.

Skills

Cloud & Platform Engineering: AWS, Azure, GCP (GKE), multi-cloud architecture, distributed systems

Kubernetes & Containers: Kubernetes (GKE/EKS/AKS), Docker, Helm, autoscaling, rolling deployments

CI/CD & DevOps: Jenkins, GitLab CI/CD, GitHub Actions, pipeline automation, JFrog

Infrastructure as Code: Terraform, Ansible, environment provisioning, automation

Programming: Python, Java, Docker, Bash, Groovy, JavaScript/TypeScript, SQL, C#, R

Observability & Reliability: Prometheus, Grafana, ELK, SLI/SLO, incident management

Experience

DevOps Engineer II, Scotiabank – Toronto, Canada

Apr 2026 – present

- Designed and operated scalable **cloud and on-prem infrastructure** supporting **enterprise data platforms** with **100+ integrated systems**
- Built and maintained **CI/CD pipelines** and **Infrastructure-as-Code (Terraform/Ansible)** to improve deployment efficiency and consistency
- Managed **Kubernetes (GKE) clusters** and containerized workloads (**Docker, Helm**) across dev, staging, and production environments
- Implemented **observability (Prometheus, Grafana, ELK)** with **SLI/SLO-driven monitoring** for proactive incident response
- Collaborated with **data engineering teams** to support large-scale data ingestion, processing, and platform reliability

Senior DevOps and Cloud Engineer, Bell Media – Toronto, Canada

Oct 2021 – July 2025

- Architected and operated **Kubernetes-based CI/CD platform** supporting large-scale distributed systems
- Designed and provisioned cloud infrastructure using **Terraform**, improving deployment consistency and scalability
- Optimized **CI/CD pipelines (Jenkins/GitLab)**, reducing build latency by **27%** through caching and workflow redesign
- Built **Python-based tooling** for automation, diagnostics, and log analysis to reduce **MTTR**
- Implemented **observability stack (Prometheus, Grafana, ELK)** for system monitoring and alerting
- Led **incident response and root cause analysis**, maintaining **>95% platform uptime**
- Standardized **artifact management** and secure software delivery using **JFrog**

AI Systems & Software Engineer (Part-time), York University – Toronto, Canada

Mar 2021 – present

- Designed and deployed **containerized AI/ML pipelines** supporting distributed model training
- Built scalable **data processing pipelines and APIs** for large-scale datasets and experimentation
- Automated research infrastructure using **Terraform and Ansible** across hybrid environments
- Developed **evaluation frameworks** with automated logging, telemetry, and benchmarking

Software Developer Intern, Bell Canada – Toronto, Canada

May 2021 – Aug 2021

- Developed **serverless applications (AWS Lambda, Glue)** for data processing and validation
- Automated **cloud-based data pipelines** improving operational efficiency and observability

Education

York University, MSc in Electrical and Computer Engineering – Toronto, Canada

Jan 2026 – Dec 2027

- Focus: **AI-driven wireless communication, LLM-assisted decision systems**, and resource-aware network optimization
- Research on **Autonomous Driving** and **6G** communication using Quantum and LLMs
- Advisor: [Prof. Hina Tabassum](#), [Prof. Ping Wang](#)

York University, BSc with Honours in Computer Science & Statistics – Toronto, Canada

Sept 2016 – Aug 2021

- Focus on AI, cloud computing, and data-intensive applications
- Capstone: Optimizing V2I Communication for Autonomous Driving using RL-based frameworks

Publications

Semantic-Aware Adaptive Video Streaming Using Latent Diffusion Models for Wireless Networks

July 2025

Zijiang Yan, Jianhua Pei*, Hongda Wu, Hina Tabassum, Ping Wang
[10.1109/MWC.001.2500068](#) (IEEE Wireless Communications)

Generalized Multi-Objective Reinforcement Learning with Envelope Updates in uRLLC-enabled Vehicular Networks

June 2025

Zijiang Yan, Hina Tabassum
[10.1109/TVT.2025.3580502](#) (IEEE Transactions on Vehicular Technology)

Hierarchical and Collaborative LLM-Based Control for Multi-UAV Motion and Communication

July 2025

Zijiang Yan, Hao Zhou*, Jianhua Pei, Hina Tabassum
[10.48550/arXiv.2506.06532](#) (ML4Wireless Workshop @ ICML 2025)

Hybrid LLM-DDQN based Joint Optimization of V2I Communication and Autonomous Driving

Feb 2025

Zijiang Yan, Hao Zhou, Hina Tabassum, Xue Liu
[10.1109/LWC.2025.3539638](#) (IEEE Wireless Communications Letters)

CVaR-Based Variational Quantum Optimization for User Association in Handoff-Aware Vehicular Networks

June 2025

Zijiang Yan, Hao Zhou, Jianhua Pei, Aryan Kaushik, Hina Tabassum, Ping Wang
[10.1109/ICC52391.2025.11161596](#) (IEEE International Conference on Communications (ICC 2025))

Optimizing Vehicular Networks with Variational Quantum Circuits-based Reinforcement Learning

May 2024

Zijiang Yan, Ramsundar Tanikella, Hina Tabassum
[10.1109/INFOCOMWKSHPS61880.2024.10620888](#) (IEEE INFOCOM 2024 Workshops)

Multi-UAV Speed Control with Collision Avoidance and Handover-Aware Cell Association - DRL with Action Branching

Dec 2023

Zijiang Yan, Wael Jaafar, Bassant Selim, Hina Tabassum
[10.1109/GLOBECOM54140.2023.10436730](#) (IEEE GLOBECOM 2023)

Reinforcement Learning for Joint V2I Network Selection and Autonomous Driving Policies Zijiang Yan , Hina Tabassum 10.1109/GLOBECOM48099.2022.10001396 (IEEE GLOBECOM 2022)	Dec 2022
Concept Drift-Aware Load Forecasting for IoT Energy Systems Leveraging TimesNet-Based Two-Layer Framework Yixiang Huang, Zijiang Yan , Jianhua Pei, Jinfu Chen (Under Review)	Mar 2026
EMFusion: Conditional Diffusion Framework for Trustworthy Frequency Selective EMF Forecasting in Wireless Networks Zijiang Yan , Yixiang Huang*, Jianhua Pei, Hina Tabassum, Luca Chiaraviglio 10.48550/arXiv.2512.15067 (Under Review)	Dec 2025

Professional Activities

Conference Session Chair: IEEE ICC 2025, IEEE GLOBECOM 2023

Technical Program Reviewer: NeurIPS, IEEE ICC, IEEE GLOBECOM, IEEE WCNC, IEEE VTC, IEEE DySPAN

Journal Reviewer: IEEE TMC, IEEE TWC, IEEE TCOM, IEEE TNSE, IEEE COMML, IEEE TNSM, IEEE WCM, IEEE IOTM

Honors and Awards

- 3rd Place – Student Innovation Competition on Sustainable Space Communications, 2025
- Bell Geekfest Inspiring Lightning Talk, 2025
- Silver Award – China Youth Creative Competition (Technology Innovation Track), 2024
- Bell Geekfest Best Presentation, 2023
- Lassonde Undergraduate Research Award (LURA), York University, 2022
- Dean's List, Lassonde School of Engineering, 2018